Intelligent Monitoring and Predicting Airborne Infection Risks in Hospital Environment Through Machine Learning Enabled IoT System

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In

COMPUTER SCIENCE AND ENGINEERING

by

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CERTIFICATE

This is to certify that the EPICS Report entitled **"Intelligent Monitoring and Predicting Airborne Infection Risks in Hospital Environment Through an Machine Learning Enabled IoT System"** being submitted by

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in partial fulfilment for the award of the Degree of **Bachelor of Technology** in **Computer Science and Engineering** to the **Jawaharlal Nehru Technological University, Kakinada**, is a record of bonafide work carried out during the A.Y 2025-26.

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DECLARATION

We hereby declare that the EPICS entitled **"Intelligent Monitoring and Predicting Airborne Infection Risks in Hospital Environment Throgh Machine Learning Enabled IoT System"** submitted for the B.Tech Degree is our original work and the dissertation has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles.

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Abstract

Infection risk prediction in indoor healthcare environments has become increasingly important due to rising concerns about airborne diseases and poor air quality conditions. Effective infection control is crucial for minimizing healthcare-associated infections (HAIs), improving patient safety, and reducing healthcare costs. Existing monitoring methods often depend on manual inspections or isolated sensor readings, which lack scalability, accuracy, and real-time responsiveness. To address these limitations, this project proposes a smart infection risk detection system using the ESP32 Wroom-32 microcontroller integrated with environmental sensors-BME280, CCS811, and GP2Y1010AU0F. These sensors monitor key parameters such as temperature, humidity, pressure, CO, VOCs, and dust concentration. The collected multi-dimensional data is analysed using the K-Means clustering algorithm to classify indoor spaces based on infection risk levels. This system offers continuous, autonomous monitoring, enabling early warnings and better management of infection-prone zones without human intervention.

Keywords: Infection risk prediction, Health care, Airborne diseases, Environmental sensors.

Chapter 1

INTRODUCTION

Hospital environments are highly sensitive spaces where maintaining optimal air quality is essential for ensuring patient safety, preventing disease transmission, and supporting healthy healthcare operations. Airborne infections, driven by factors such as increased particulate matter, elevated humidity, poor ventilation, and high levels of volatile organic compounds (VOCs), can significantly increase the risk of Hospital-Acquired Infections (HAIs). These infections create additional burdens for patients, healthcare staff, and the entire medical system. With modern hospitals growing more crowded and environmental conditions fluctuating continuously, manual monitoring methods are no longer sufficient to maintain air safety.

To address this challenge, advancements in the Internet of Things(IoT) and Machine Learning (ML) offer new opportunities for intelligent, automated environmental monitoring.IoT devices equipped with sensors enable continuous data collection of critical parameters such as temperature, humidity, CO concentration, VOC levels,and particulate matter. When combined with ML algorithms such as K-Means clustering, these systems can categorize real-time environmental conditions and predict infection risk levels without requiring human intervention. In this project, we propose a smart infection-risk prediction system using an ESP32 microcontroller integrated with environmental sensors-BME280 , CCS811 , and GP2Y1010AU0F. By applying a Transformer-based pipeline for data processing and K-Means clustering for unsupervised classification, the system identifies low, medium, and high-risk zones in hospital environments. This automated solution enables healthcare staff to take preventive actions promptly, making infection control more effective and reliable.

1.1 Basic Concepts

1.1.1 Airborne Infection Risks

Airborne infection risks refer to the possibility of disease transmission through particles or droplets present in the air. These particles may include dust, pathogens, aerosols, and other contaminants that can linger in enclosed indoor spaces, especially in hospitals where patients with communicable diseases may be present. Environmental factors such as temperature, humidity, CO concentration, and particulate levels directly influence how long airborne pathogens survive and how easily they spread. By continuously measuring these parameters, hospitals can identify hazardous conditions early and implement corrective actions to minimize infection risks.

1.1.2 IoT-Based Environmental Monitoring

IoT-based environmental monitoring involves using embedded sensors connected to microcontrollers to collect real-time data on environmental conditions. In healthcare settings, IoT sensors measure key air parameters like humidity, VOCs, dust levels, and atmospheric pressure. This data is transmitted to cloud platforms such as ThingSpeak for analysis and visualization. IoT systems improve hospital safety by offering constant surveillance, immediate alerts, and automated decision support, thereby reducing human workload and enabling faster responses to environmental hazards.

1.1.3 Machine Learning for Risk Prediction

Machine Learning (ML) enables automated analysis of high-dimensional sensor data to uncover patterns not easily visible through manual monitoring. In this project, unsupervised ML-specifically, the K-Means clustering algorithm is used to group environmental conditions into different risk levels without requiring labeled data. K-Means identifies clusters representing low, medium, and high infection risks based on similarities in temperature, humidity, VOCs, dust concentration, and CO levels.

1.1.4 ESP32 Microcontroller

The ESP32 is a powerful microcontroller widely used in IoT applications due to its built-in Wi-Fi, Bluetooth, and efficient processing capabilities. It supports real-time data acquisition from sensors, preprocessing of environmental data, and wireless data transmission to cloud platforms. The ESP32's dual-core architecture and low energy consumption make it suitable for continuous hospital monitoring systems. In this project, it acts as the central processing unit that gathers sensor data, applies ML computation where needed, and sends updates to the cloud for further analysis.

1.1.5 Environmental Sensors

Environmental sensors play a crucial role in detecting the factors associated with airborne infection risks:

- **BME280:** Measures temperature, relative humidity, and atmospheric pressure.
- **CCS811:** Detects total VOC concentrations and estimates CO levels.
- **GP2Y1010AU0F:** Measures particulate matter (dust) concentration.

These sensors collectively offer a detailed examination of indoor air conditions and provide the data required for accurate risk prediction.

1.2 Motivation

The motivation behind this project stems from the growing concern over Hospital-Acquired Infections (HAIs), which continue to pose a significant threat to patient health worldwide. Many hospitals still rely on manual inspections or threshold-based systems that fail to detect early signs of hazardous environmental changes. Furthermore, rural hospitals and small healthcare centers often lack advanced equipment to continuously monitor air quality, increasing the likelihood of infection spread.

The need for a cost-effective, automated, and intelligent monitoring system is therefore crucial. By leveraging IoT sensors and ML techniques, this project aims to provide

hospitals with a smart, real-time infection-risk detection tool. The system empowers medical staff with instant notifications, better decision-making support, and improved infection control strategies-ultimately enhancing patient safety and reducing healthcare complications.

1.3 Problem Statement

Hospitals face increasing challenges in maintaining safe indoor air quality due to varying patient density, poor ventilation, and changing environmental conditions. Traditional monitoring systems are limited by manual checks, delayed reporting, and the inability to detect complex interactions among air quality parameters. These limitations contribute to frequent outbreaks of airborne infections, increasing medical costs and patient risks.

The problem addressed in this project is the lack of an automated, real-time, ML-driven system for predicting airborne infection risks in hospital environments. The objective is to build an IoT-based solution that continuously collects environmental data through sensors, processes it using ML models, and classifies infection risk levels. This system must be low-cost, accurate, scalable, and capable of generating timely alerts to support preventive actions by hospital staff.

1.4 Scope

1. Continuous monitoring of environmental parameters such as temperature, humidity, VOCs, CO, and dust levels inside hospital rooms using IoT sensors.

2. Classification of infection risk levels (Low, Medium, High) using ML-based clustering algorithms based on real-time sensor data.

3. Integration of cloud platforms like ThingSpeak for data storage, visualization, and automated alerting.

4. Deployment of an automated notification system to inform hospital management when risks are detected.

1.5 Objectives

1. To analyze existing research on air quality monitoring and identify gaps in current hospital infection control systems.

2. To design and develop an IoT-based device using ESP32 and environmental sensors to collect real-time air quality data.

3. To train and implement a K-Means clustering model for predicting infection risk levels based on sensor data.

4. To integrate the ML model with cloud platforms and automate alert notifications for medium and high-risk levels.

5. To evaluate system performance by testing the prototype in a controlled environment and assessing reliability, accuracy, and speed.

1.6 Advantages

1. Automated and real-time detection of airborne infection risks using IoT and ML-based analysis.

2. A low-cost and scalable solution suitable for hospitals of all sizes, including remote and resource-constrained facilities.

3. Continuous data logging, visualization, and instant alert notifications through cloud integration.

4. Reduces dependency on manual inspections and improves the speed of preventive actions.

5. Enhances patient safety by providing early detection of infection-prone environmental conditions.

Chapter 2

LITERATURE REVIEW

2.1 Low-Cost IoT-Based Indoor Air Quality Monitoring System for Healthcare Environments [7]

Journal: Frontiers in Sustainable Cities, 2022

Methodology:

Abubeker et al. [7] from the School of Computer Science, Addis Ababa University, Ethiopia, in 2022 developed a low-cost indoor air quality monitoring system aimed at minimizing non-ventilator hospital-acquired pneumonia (NVHAP). Their system integrated an ESP8266 microcontroller with environmental sensors such as MQ-135 (CO_2, NH_3), HC-SR04 ultrasonic sensor, and moisture sensors. Data were transmitted via a wireless body sensor network (WBSN) for real-time alerts. Threshold-based analysis was used to detect unsafe air conditions.

Advantages:

1. Low-cost solution suitable for resource-limited hospitals.
2. Enabled remote monitoring through a wireless sensor network.
3. Sensors captured diverse environmental parameters.

Disadvantages:

1. Lacked PM2.5 and VOC detection.
2. No machine learning implementation for predictive analysis.

2.2 IoT-Based Air Quality and Dust Monitoring System for Hospitals [8]

Journal: Discover Internet of Things, 2025

Methodology:

Parkavi et al. [8] from SNS College of Technology developed an IoT-based air quality system using Arduino UNO, MQ135, GP2Y1010AU0F dust sensor, and DHT11. Sensor data were visualized through a ThingSpeak-integrated mobile app. Alerts were triggered when parameters exceeded safe thresholds.

Advantages:

1. Supported threshold-based real-time alerts.
2. Cloud integration enabled remote monitoring.
3. User-friendly application interface.

Disadvantages:

1. Frequent calibration required.
2. Slow 120-second update interval reduces responsiveness.

2.3 Air Quality Monitoring and Control System for Hospital Rooms [9]

Journal: International Conference on e-Health and Bioengineering, 2021

Methodology:

Andrei et al. [9] proposed an Arduino-based indoor air monitoring system detecting temperature, humidity, and CO_2, integrated with Bluetooth-based mobile visualization. Automatic ventilation was triggered when unsafe conditions occurred.

Advantages:

1. Simple design, easy for rapid deployment.
2. Automatic ventilation assists healthcare staff.
3. Low power consumption.

Disadvantages:

1. Lacked PM2.5 detection.
2. No VOC measurement for complete analysis.

2.4 Fog and Cloud-Based Hospital Air Quality Monitoring System [10]

Journal: International Conference on Cybernetics and Intelligent Systems, 2022

Methodology:

Ahmadi et al. [10] developed an ESP32-based IoT system using fog and cloud computing layers for low-latency air quality monitoring. PM2.5, MQ135, and DHT sensors were used. Alerts were delivered through Telegram and buzzer notifications.

Advantages:

1. Very fast alerting due to fog computing.
2. Hybrid cloud-fog design improves reliability.

Disadvantages:

1. Heavy dependency on internet connectivity.
2. Lacked VOC monitoring.

2.5 Indoor Air Quality Monitoring System for Healthcare Using Arduino [11]

Journal: IEEE ICONSTEM, 2024

Methodology:

Kumar et al. [11] implemented an Arduino-based air quality system using MQ135 sensors to measure CO_2 and VOCs. Data were displayed locally and alerts were issued via LEDs and buzzers.

Advantages:

1. Low-cost design for small healthcare centers.
2. Real-time local alerts.

Disadvantages:

1. No PM2.5 detection.
2. No cloud integration for remote monitoring.

2.6 Environmental Health Monitoring System Using IoT and ML [12]

Journal: International Conference on Advances in Computing, Communication and Networking, 2024

Methodology:

Chauhan et al. [12] developed an IoT–ML system using multiple sensors with K-Means clustering and logistic regression to classify environmental states. Cloud dashboards were used for visualization.

Advantages:

1. Integrated IoT and ML for predictive analytics.
2. Cloud-based visualization for remote monitoring.

Disadvantages:

1. Needed large training datasets.
2. High power consumption due to multiple sensors.

2.7 Real-Time Environmental Monitoring Using ESP32 for Smart Hospitals [13]

Journal: Procedia Computer Science, 2024

Methodology:

Espinosa-Gavira et al. [13] evaluated ESP32 performance for real-time synchronized air-quality monitoring. Tests showed accurate multi-sensor synchronization and low latency across wireless networks.

Advantages:

1. High-speed Wi-Fi data transmission.
2. Dual-core ESP32 supports parallel sensor processing.

Disadvantages:

1. Requires stable Wi-Fi connectivity.

2. Limited onboard memory for heavy ML models.

Chapter 3

DESIGN THINKING

3.1 Problem Statement

Ravi, a 22-year-old patient admitted to a district hospital, is recovering from a respiratory illness. However, due to poor ventilation, high humidity levels, and elevated dust concentration inside the general ward, he becomes increasingly vulnerable to airborne infections. The hospital staff rely on manual inspections to monitor air quality, but these checks are infrequent, inconsistent, and often fail to detect hazardous conditions in time. The lack of real-time monitoring tools prevents early identification of infection-prone environments. With limited resources and outdated systems, hospitals cannot continuously track critical environmental parameters such as CO_2, VOCs, temperature, humidity, and particulate matter. An IoT and ML-powered device can bridge this gap by offering continuous monitoring, automated risk prediction, and timely alerts. Accessible and low-cost, this solution empowers hospitals to detect infectious conditions early, improving patient safety and reducing the spread of airborne diseases.

3.2 Primary Research

We conducted a survey among hospital staff, patients, and attendants to understand challenges related to indoor air quality and infection risk. The following questions were included:

1. Are you aware that poor indoor air quality can increase infection risks in hospitals?
2. Have you experienced discomfort due to temperature, dust, or odors inside hospital wards?

3. Do you believe hospitals should monitor air quality in real time?

4. What factors do you think contribute most to airborne infection spread?

5. Would automated alerts about infection-prone conditions help hospital staff take preventive actions?

SURVEY RESULTS

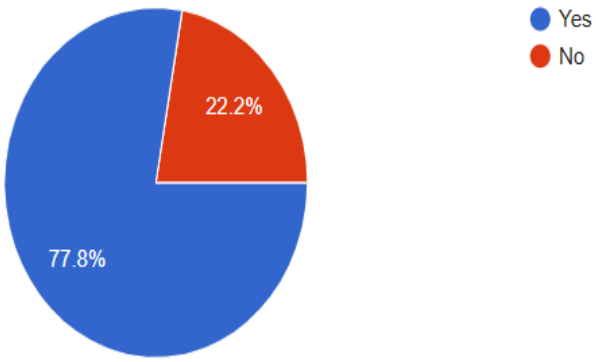


Figure 3.1: Awareness of indoor air quality and infection risks

Figure 5.1 shows that only 22% of respondents were aware that air quality affects infection risk, while nearly 77% were unaware, indicating a significant knowledge gap that increases infection exposure in hospitals.

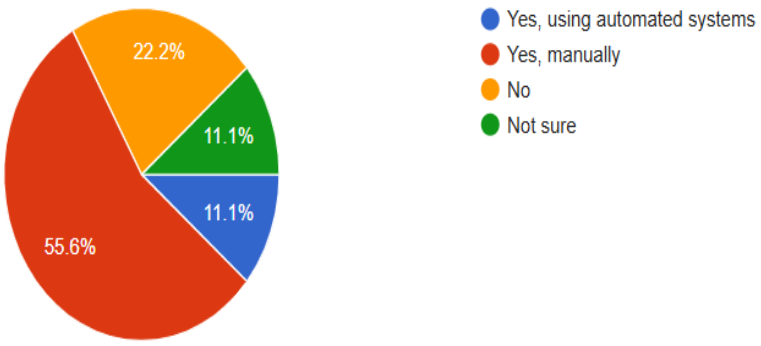


Figure 3.2: Preferences for monitoring and alert mechanisms

Figure 5.2 illustrates that 11% of respondents prefer automated alert systems, 11% prefer visual dashboards, and the 55% manual logs or periodic checks and 22% others.

3.3 Secondary Research

In secondary research, we reviewed journals, articles, and conference papers related to IoT-based air quality monitoring and infection prediction.

Table 3.1: Secondary research for IoT-based infection risk prediction system

S.No	Author	Title	Advantages	Disadvantages
1	Abubeker KM et al. [7]	Wireless sensor-based air quality monitoring system in hospitals	Low-cost hardware; Wireless monitoring minimized manual effort; Multiple sensors improved coverage.	No PM2.5/VOC detection; No predictive ML analysis.
2	Parkavi A. et al. [8]	IoT-based air quality and dust level monitoring in hospitals	Real-time alerts; Cloud dashboards; User-friendly mobile interface.	Frequent calibration; 120-sec update delays risk detection.
3	Ahmadi C. et al. [9]	Fog and cloud computing-based air quality monitoring	Fast alerts using fog computing; Multi-channel notification; Hybrid storage.	Network dependency; No CO/VOC monitoring.

S.No	Author	Title	Advantages	Disadvantages
4	Espinosa-Gavira M. et al. [10]	Characterization of ESP32 for real-time IoT networks	Low latency; High efficiency; Dual-core ideal for IoT.	Insufficient memory for heavy ML; Wi-Fi dependent.
5	Chauhan S. et al. [11]	Environmental health monitoring using IoT and ML	Integrated K-Means; Cloud visualization; Real-time alerts.	ML requires high-quality datasets; Higher power usage.

3.4 Ideation Phase

Based on insights from the Empathy phase, the problem was defined as the lack of continuous environmental monitoring and predictive analysis in hospital wards. Priority was given to developing a solution that ensures accuracy, affordability, and real-time detection. Proposed ideas included:

- Deploying an IoT device with multiple environmental sensors to monitor air quality parameters continuously.
- Implementing machine learning algorithms such as K-Means clustering to categorize infection risk levels.
- Using cloud platforms (ThingSpeak) to store and visualize data and enable automated notifications.
- Designing alert systems (email/telegram) to inform staff immediately when risk levels exceed safety thresholds.
- Adding a dashboard to help hospital administrators track historical trends and improve decision-making.

3.5 Solution and Prototype Phase

Our solution includes developing a smart IoT device built on the ESP32 microcontroller integrated with sensors such as BME280, CCS811, and GP2Y1010AU0F. This system collects real-time air quality data and uploads it to ThingSpeak using Wi-Fi. A machine learning model based on K-Means clustering is linked to the cloud to predict infection risks. When medium or high-risk levels are detected, notifications are automatically sent to healthcare staff via telegram. The prototype focuses on continuous monitoring, autonomous risk detection, and reliable alert mechanisms.

3.6 Prototyping

Prototyping involved assembling the ESP32 with environmental sensors on a breadboard, writing firmware for sensor calibration and data collection, and connecting the device to the ThingSpeak IoT cloud. The collected data was preprocessed and used to train the K-Means clustering model. Using MATLAB Analysis on ThingSpeak, the model was integrated to run at fixed intervals for real-time risk classification. The prototype was tested in controlled indoor conditions to validate sensor accuracy and model performance.

3.7 Final Solution

The final solution is a fully integrated infection risk monitoring system capable of collecting, analyzing, and classifying air quality data in real time. It consists of:

- An ESP32-based IoT device continuously recording temperature, humidity, pressure, VOCs, CO_2, and dust levels.
- A trained K-Means ML model deployed on ThingSpeak.
- Automatic alerts sent via telegram when risk levels reach medium or high.
- A cloud dashboard that visualizes sensor readings and cluster predictions.

This solution enables early detection of infection-prone conditions, helping hospital staff take preventive measures before risks escalate.

3.8 Challenges

Developing this system presented several challenges. Ensuring accurate readings required frequent sensor calibration and noise filtering. The ML model needed high-quality, well-preprocessed data to classify risk levels reliably. Wi-Fi connectivity issues occasionally interrupted real-time updates. Deploying the ML model on a cloud platform required optimization to handle continuous data flow without delays. Finally, balancing low cost with high accuracy demanded careful hardware selection and efficient power usage.

Chapter 4

SOFTWARE REQUIREMENTS ANALYSIS

Requirements analysis is a crucial stage that determines the success of any system development process. Requirements are categorized into two major types: functional and non-functional requirements. For the proposed Hospital Infection Risk Detection System, analyzing these requirements ensures that the system performs accurately, efficiently, and consistently in real-world healthcare environments.

4.1 Functional Requirements

Functional requirements specify the core functionalities and operations the system must perform to meet user needs. These functions describe how the system should behave under various conditions, detailing the interactions between the sensors, microcontroller, cloud, machine learning model, and end users. Functional requirement analysis includes steps such as sensor integration, data acquisition, model processing, and real-time alerting. Table 4.1 outlines the ranked functional requirements of the system.

Table 4.1: Functional Requirements for Infection Risk Prediction System

Req. ID	**Requirement**	**Must/Want**	**Comments**
FR001	Real-time Sensor Data Acquisition	Must	ESP32 must continuously collect data from BME280, CCS811, and Dust Sensor.
FR002	Preprocessing of Environmental Data	Must	Sensor readings must be filtered, calibrated, and validated before cloud upload.
FR003	Cloud Data Uploading	Must	ESP32 must send processed data to ThingSpeak for dashboards and ML.
FR004	K-Means++ Clustering for Risk Prediction	Must	System must classify environment into Low, Medium, and High risk levels.
FR005	Automated Alert Notification	Must	Notification alerts must be sent when infection risk exceeds safe thresholds.
FR006	Local Display of Readings	Want	LCD/OLED may optionally show real-time sensor status.
FR007	Data Logging	Want	Historical sensor data may be stored for audits and long-term analysis.

4.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system. They affect system performance, reliability, scalability, and ease of use. These requirements are essential to ensure the infection risk prediction system remains efficient and trustworthy in healthcare environments.

Table 4.2: Non-Functional requirements for infection risk prediction system

Requirement ID	**Requirement Statement**	**Must/ Want**	**Comments**
NF001	High Accuracy in Risk Prediction	Must	Ensures correct classification of environmental conditions to protect patient safety.
NF002	System Scalability	Must	Should support additional sensors, rooms, or hospital wards without redesign.
NF003	Low Response Time	Must	The model must evaluate and trigger alerts within seconds to maintain real-time monitoring.
NF004	System Reliability	Must	Device should operate continuously with minimal downtime, especially in critical care units.
NF005	Data Security and Integrity	Must	Protects sensor data and hospital staff information during cloud transmission.
NF006	Maintainability	Want	System updates (sensor calibration, ML model tuning) should be easy with minimal disruption.
NF007	Low Power Consumption	Must	Ensures long-term operation in hospital environments with minimal energy usage.

Non-functional requirements set constraints on the architecture, design, and deployment of the system across various environments. They ensure operational efficiency, reliability, and user satisfaction. Table 4.2 summarizes the major non-functional requirements essential for implementing the Hospital Infection Risk Detection System.

4.3 System Requirements

System requirements include hardware, software, and operational specifications needed to run the infection risk prediction system effectively. These requirements ensure optimal system performance, seamless integration, and compatibility across different environments.

4.3.1 Software Requirements

The system requires specific libraries, firmware, and development tools for implementation. The required software includes Arduino IDE, Python, ThingSpeak API, MATLAB Analysis, and ML libraries. These tools establish the environment for coding, cloud integration, data processing, and ML model execution.

4.3.1.1 Arduino IDE

Arduino IDE is used for programming the ESP32 microcontroller. It supports sensor drivers, Wi-Fi integration, and serial communication. It offers a straightforward development environment for IoT applications.

4.3.1.2 Python

Python is used to preprocess data, train the K-Means clustering model, and visualize experimental outputs. It is widely used for numerical computing and ML algorithms.

4.3.1.3 MATLAB (ThingSpeak)

MATLAB Analysis on ThingSpeak is used to execute the K-Means clustering code in the cloud. It enables real-time ML execution triggered by new sensor data.

4.3.1.4 ThingSpeak API

ThingSpeak acts as an IoT analytics platform where sensor data is stored, visualized, and processed. It supports MATLAB scripts, triggers, and alert mechanisms.

4.3.1.5 ESP32 Libraries

The system requires various Arduino libraries such as:

- **WiFi.h** - Manages ESP32 Wi-Fi communication
- **Adafruit_BME280.h** - Reads temperature, humidity, pressure
- **CCS811.h** - Reads VOCs and CO_2
- **Wire.h** - I2C communication

4.3.1.6 MQTT / HTTP Protocol Libraries (Optional)

Supports alternative ways for ESP32 to send data to remote servers.

4.3.2 Hardware Requirements

The hardware components required for developing and deploying the infection risk system include:

- **Microcontroller:** ESP32 with integrated Wi-Fi.
- **Sensors:**
  - BME280 - Temperature, Humidity, Pressure
  - CCS811 - VOCs and CO_2
  - GP2Y1010AU0F - Dust / Particulate Matter
- **Power Supply:** 5V/2A adapter or power bank.
- **Connectivity:** Stable Wi-Fi for real-time cloud updates.
- **Prototype Components:** Breadboard, jumper wires, resistors, USB cable.
- **Optional:** OLED/LCD display for on-device monitoring.

These hardware components ensure smooth data acquisition, real-time cloud communication, and accurate ML-based infection risk prediction.

Chapter 5

SOFTWARE DESIGN

5.1 Software Development Lifecycle

Figure 5.1 illustrates the lifecycle model adopted for the proposed Hospital Infection Risk Detection System. The Scrum framework, a widely used Agile methodology, was selected to manage the project efficiently. Scrum enables rapid adaptation to changing requirements, supports iterative development, and ensures continuous improvement through collaborative teamwork. Its flexibility makes it suitable for IoT and machine learning projects where real-time feedback and incremental enhancements are essential.

The product backlog includes all essential tasks such as sensor integration, ESP32 firmware development, cloud configuration, machine learning model training, alert generation, and dashboard development. During sprint planning, the team selects the priority items for the sprint and establishes the sprint goal focused on functional increments. A sprint typically lasts 2-3 weeks, during which daily stand-up meetings enable the team to track progress, discuss blockers, and refine plans.

At the end of each sprint, the sprint review showcases the completed work—including sensor data acquisition, cloud dashboards, ML clustering results, and telegram alert functionality - to stakeholders for feedback. The Sprint Retrospective helps the team evaluate what worked well, what challenges arose, and what improvements can be implemented in the next sprint.

Scrum involves essential roles: the Scrum Master ensures smooth execution and removes impediments, while the Product Owner prioritizes backlog items based on clinical importance. Artefacts such as the Burndown Chart track progress, and the Definition of Done ensures each feature meets the project-quality standards before being accepted as an increment.

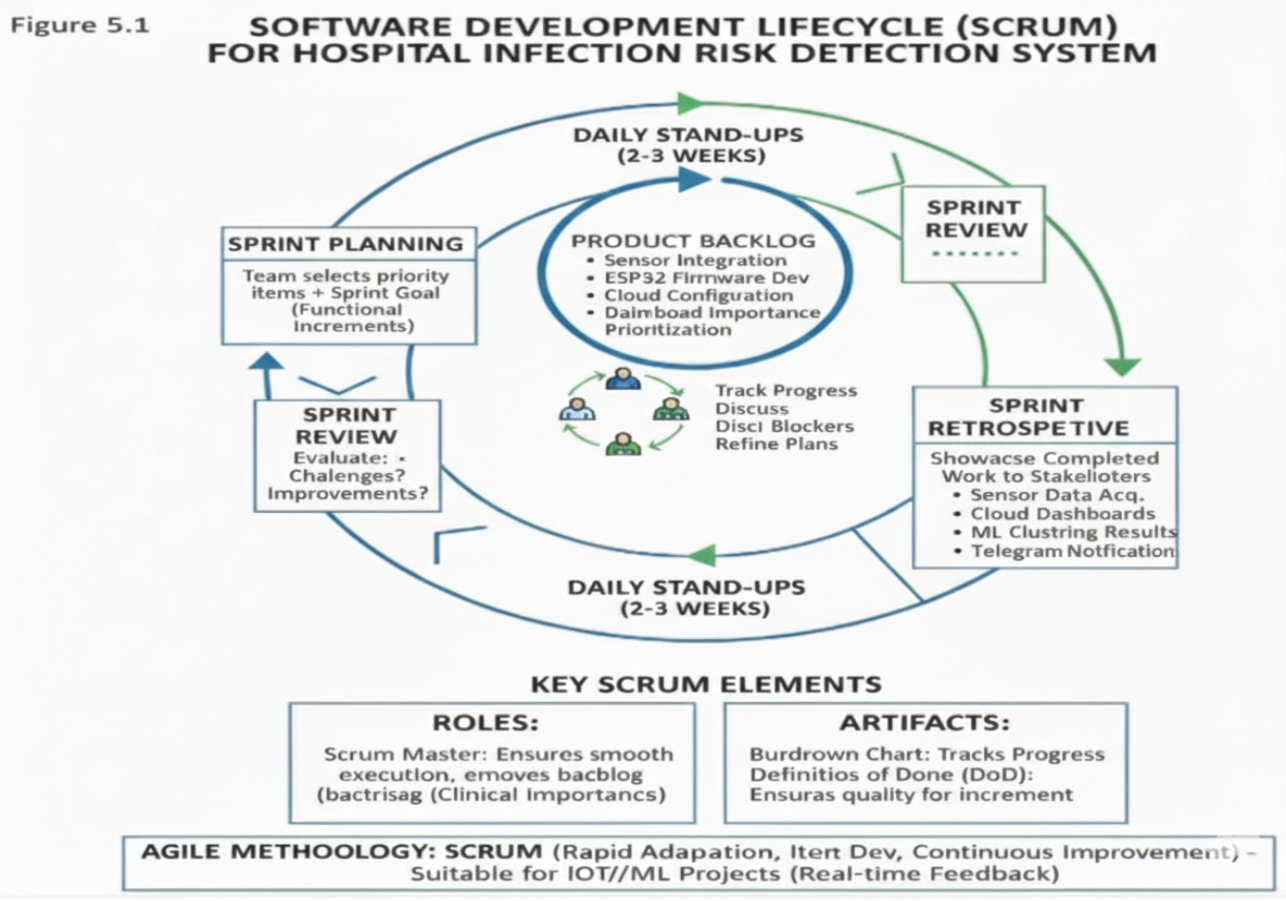


Figure 5.1: Software Development Lifecycle (SCRUM) for Infection Risk Detection System[21]

5.2 UML Diagrams

Unified Modeling Language (UML) provides a standardized way to visualize the architecture and workflow of complex systems. For the Infection Risk Detection System, UML diagrams help represent user interactions, system processes, sensor workflows, machine learning processing, and cloud operations.

5.2.1 Use-Case Diagram

A Use Case diagram visually represents interactions between the system and external users or devices. In this project, the primary actors are the Hospital Staff, IoT Device (ESP32), Sensor Modules, and Cloud Server (ThingSpeak). Figure 5.2 illustrates the Use Case diagram.

The diagram includes functionalities such as monitoring environmental parameters,

uploading data to the cloud, running ML risk prediction, accessing historical reports, and sending automated alerts. Hospital Staff can check real-time data and receive warnings when infection risk is elevated.

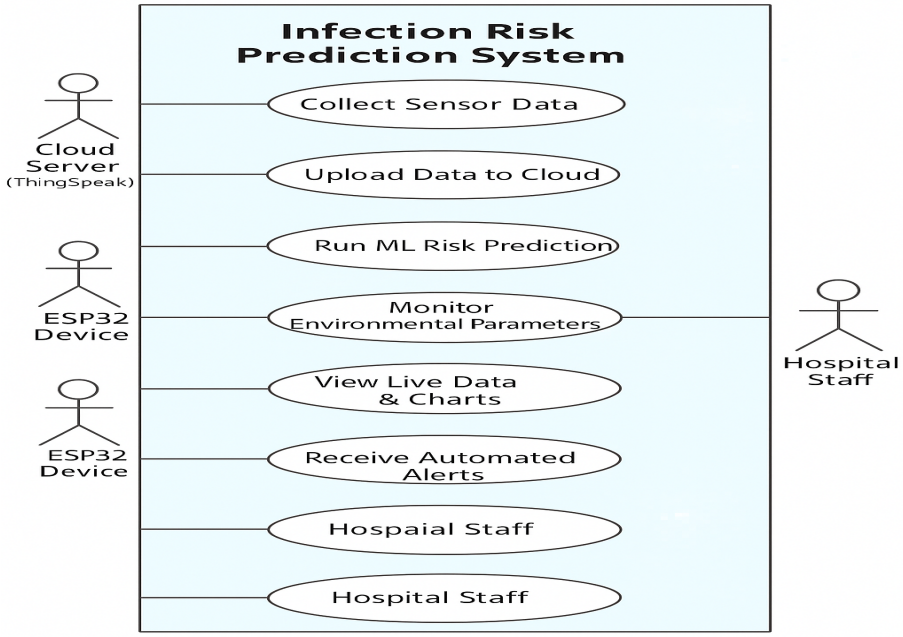


Figure 5.2: Use Case diagram of Infection Risk Detection System[21]

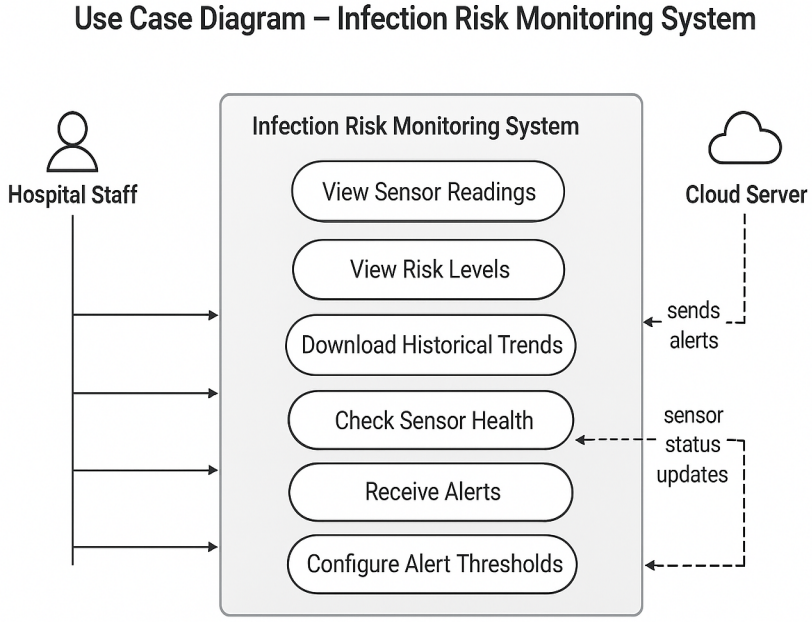


Figure 5.3: Use Case diagram illustrating user activities in the system[21]

The Use Case diagram in Figure 5.3 elaborates on activities such as viewing risk levels, downloading historical trends, checking sensor health status, and configuring alert thresholds.

5.2.2 Activity Diagram

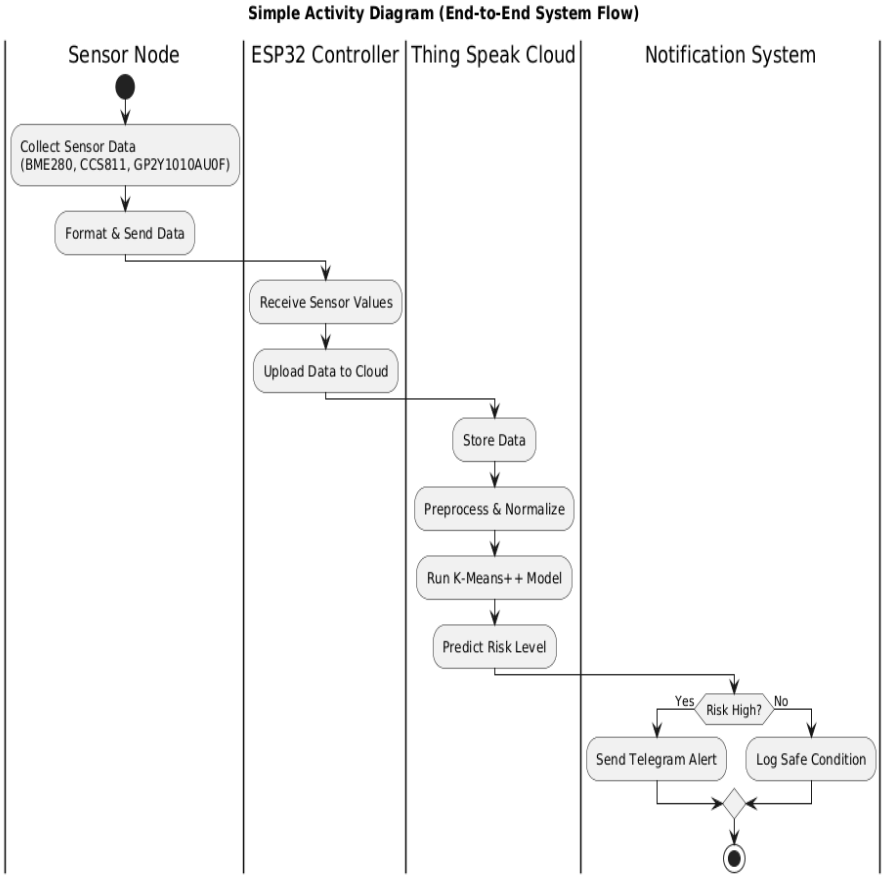


Figure 5.4: Activity diagram of Infection Risk Detection System IoT workflow[22]

Figure 5.4 describes the complete flow: sensor data collection, ThingSpeak upload, ML inference, and alert notification generation when the predicted risk is Moderate or High

5.2.3 Sequence Diagram

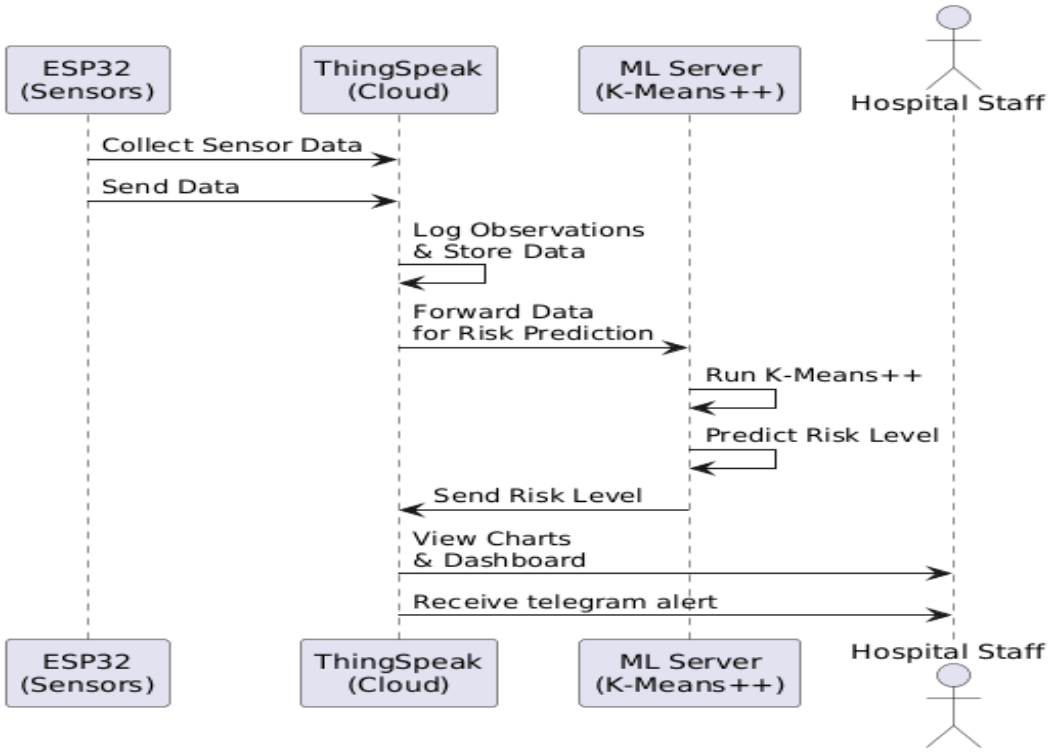


Figure 5.5: Sequence diagram showing ESP32 - Cloud - user interaction[22]

Figure 5.5 shows how the ESP32 sensor node collects environmental data and sends it to the ThingSpeak cloud for logging and storage. The cloud forwards the data to the ML server, where the K-Means++ model performs infection-risk prediction. The predicted risk level is then returned to the cloud, which updates the dashboard and alerts the hospital staff if the risk is high. This flow shows the complete interaction between the device, cloud, ML model, and end user.

Chapter 6

PROPOSED SYSTEM

6.1 Proposed System Architecture

The primary objective of the proposed system architecture is to build a reliable, real-time Infection Risk Detection System capable of monitoring environmental conditions inside hospitals and predicting infection-prone zones. The system emphasizes scalability, adaptability, accuracy, and continuous monitoring by integrating IoT-based sensing, cloud storage, and machine learning analysis. The Process Flow Diagram, shown in Figure 6.1, represents the sequential flow of operations from sensor data acquisition to ML-based risk prediction and alert generation.

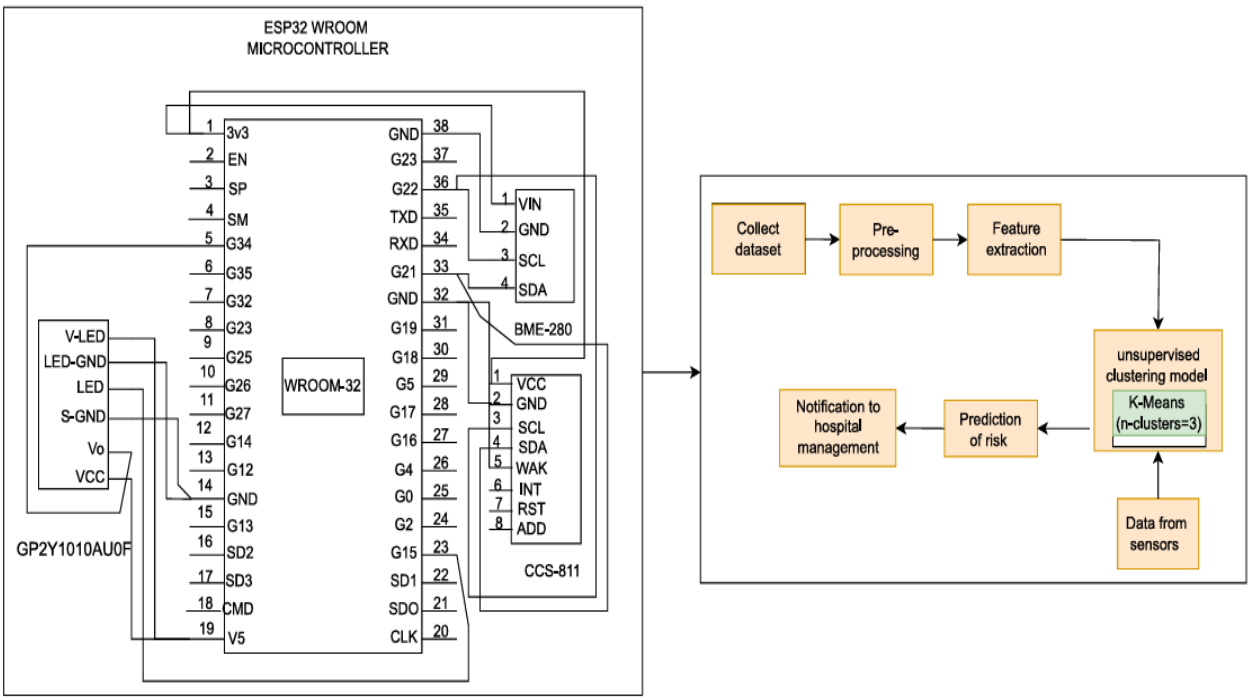


Figure 6.1: Proposed System Architecture diagram of the infection risk detection system.

The architecture depicted in Figure 6.1 outlines the overall structure of the Infection Risk Detection System. The system operates using an ESP32 microcontroller connected to multiple environmental sensors such as the BME280 (temperature, humidity, pressure), CCS811 (CO2, VOCs), and GP2Y1010AU0F (dust concentration). These sensors continuously monitor environmental conditions known to influence infection spread in healthcare settings.

The sensed data is processed by the ESP32 and sent to the cloud platform ThingSpeak using Wi-Fi. ThingSpeak stores the data, generates visual dashboards, and forwards the dataset for machine learning analysis. The ML module, implemented using K-Means clustering, classifies the area into Low, Medium, or High infection risk based on sensor readings. The results are sent back to ThingSpeak, where automated alerts (e.g., telegram notifications) are triggered when risk crosses predefined thresholds.

In the initial phase, all sensor readings from hospital rooms or test environments are collected and stored in CSV format. This dataset undergoes data preparation, including cleaning, normalization, and feature extraction. The processed data is then used to train the K-Means clustering model.

Before training, the dataset is divided into training and testing samples. After training, the system performs performance evaluation, validates cluster formations, and applies parameter tuning if required.

Once the model achieves stable clusters and accurate risk grouping, it is deployed to the cloud. The ESP32 device sends continuous live readings, which are processed by the ML model to generate a risk score. The classification output is returned to the cloud service and displayed to the user through the dashboard. If infection risk levels become critical, automated alerts notify healthcare staff in real time.

6.2 Methodology

6.2.1 ESP32 Microcontroller

The ESP32 serves as the central processing and communication unit of the proposed system. It features a dual-core processor, integrated Wi-Fi and Bluetooth, and multiple GPIO pins, making it ideal for IoT-based monitoring. The ESP32 reads analog and digital data from sensors, preprocesses it, and sends it to ThingSpeak using HTTP protocols. Its low-power consumption and robust real-time performance make it suitable for continuous hospital environment monitoring.

6.2.2 Sensor Integration

The Infection Risk Detection System uses multiple sensors to capture environmental conditions associated with airborne infections:

- **BME280 Sensor:** Measures temperature, humidity, and atmospheric pressure.
- **MQ-Series Sensors:** Detect harmful gases, pollutants, and overall air quality.
- **DSM501A Dust Sensor:** Captures dust density levels that correlate with microbial spread.

Each sensor continuously collects data and sends it to the ESP32, which performs preliminary filtering before cloud upload.

6.2.3 IoT Cloud Integration

ThingSpeak is used as the cloud IoT analytics platform. It receives data from the ESP32 at predefined intervals and stores it in dedicated channels. Features such as real-time plotting, MATLAB analytics, and event-based triggers allow efficient monitoring.

The cloud platform also integrates the ML model, which processes historical datasets to produce infection risk predictions.

6.2.4 Machine Learning Model: K-Means Clustering

K-Means clustering is used as the machine learning technique for classifying infection risk. It groups environmental data into three distinct clusters:

- **Cluster 0: Low Risk**
- **Cluster 1: Medium Risk**
- **Cluster 2: High Risk**

The ML methodology involves the following steps:

6.2.4.1 Data Preprocessing

Collected sensor data is cleaned, normalized, and converted into a structured CSV dataset. Features such as temperature, humidity, gas concentration, and dust density are selected for clustering.

6.2.4.2 Model Training

K-Means is trained with historical data to form meaningful clusters. The elbow method and silhouette score are used to validate cluster quality.

6.2.4.3 Model Testing

New unseen data is fed into the trained model to evaluate prediction accuracy and cluster consistency.

6.2.4.4 Risk Classification

Each new sensor reading is classified into a risk category in real time.

6.2.5 System Deployment

After successful training and testing, the model is deployed as part of the ThingSpeak cloud environment. Deployment consists of three major stages:

1. **Embedding the ML script into ThingSpeak MATLAB Analytics:** The K-Means trained model is integrated using MATLAB code to ensure automated predictions.
2. **Linking the IoT device with the ML backend:** The ESP32 continuously uploads data, which is processed by ThingSpeak triggers to generate predictions.
3. **Alert Generation:** When risk crosses a predefined threshold, the system sends automated telegram alerts to healthcare staff.

Post-deployment, the system undergoes continuous monitoring to ensure prediction accuracy. The model is periodically updated using new data to maintain reliability. Its real-time nature ensures hospitals can respond promptly to increasing infection threats, aiding in infection control and improving patient safety.

Chapter 7

RESULTS AND DISCUSSION

7.1 Data Collection

Developing an Infection Risk Detection System requires a sufficiently large and diverse environmental dataset. Continuous sensor readings were collected using the BME280 (temperature, humidity, pressure), CCS811 (eCO_2, TVOC), and GP2Y1010AU0F (dust density) sensors interfaced with the ESP32 microcontroller. Data was streamed to the ThingSpeak cloud platform and later exported in CSV format.

The raw dataset underwent preprocessing, including noise removal, normalization, handling missing values, and feature selection. Risk labels (Low, Medium, High) were assigned using environmental threshold guidelines to support unsupervised clustering evaluation.

7.2 Results and Analysis

To determine the optimal clustering configuration, K-Means and K-Means++ were applied across three versions of the dataset:

1. Original 6-dimensional dataset
2. PCA-reduced 2-dimensional dataset
3. PCA-reduced 3-dimensional dataset

A detailed comparison is shown in Table 7.1.

Table 7.1: Comparison of K-Means++ and K-Means on 6D, PCA-2D, and PCA-3D Data

Model	**Inertia**	**Silhouette Score**	**Cluster Distribution**
K-Means++ (6D original)	6233.86	0.4291	{0: 2037, 1: 5, 2: 1}
K-Means++ (PCA 2D)	**2562.99**	**0.63296**	{0: 701, 1: 644, 2: 755}
K-Means++ (PCA 3D)	4707.53	0.47718	{0: 1027, 1: 1011, 2: 5}

Interpretation of Results

1. **K-Means++ (6D original) Analysis**

   Although the algorithm produced a moderate silhouette score (0.4291), the cluster distribution was extremely imbalanced (2037 samples in one cluster). This indicates poor separation in the original feature space.

2. **K-Means++ (PCA 3D) Analysis**

   The PCA 3D model improved cluster separation slightly but resulted in a silhouette score of 0.47718. The distribution remained skewed, with one dominant cluster and one very small cluster.

3. **K-Means++ (PCA 2D) - Final Selected Model**

   This configuration achieved the **best overall clustering performance**, with a silhouette score of 0.63296 and balanced cluster distribution (701, 644, 755). PCA 2D significantly improved separability while reducing noise, making this the final selected model.

- **Silhouette Score: 0.63296 (highest)**
- **Balanced cluster distribution: 701, 644, 755**
- **Lowest inertia compared to other models**

The PCA 2D transformation enhanced cluster separability, reduced noise, and preserved meaningful structure in the data, resulting in clear Low, Medium, and High infection risk groups.

Table 7.2: Final Selected Model: K-Means++ with PCA 2D

Metric	Value
Inertia (WCSS)	2562.99
Silhouette Score	0.63296
Davies–Bouldin Index	0.37296
Calinski–Harabasz Score	2191.85
Cluster Distribution	{0: 701, 1: 644, 2: 755}

Why PCA 2D K-Means++ Was Selected

The PCA 2D K-Means++ model provided:

- **Highest cluster separation** (Silhouette)
- **Lowest cluster overlap** (DBI)
- **Best structural definition** (CH Score)
- **More balanced and meaningful cluster counts**

These characteristics make K-Means++ (PCA 2D) the most reliable for real-time infection risk prediction.

7.3 Performance Metrics

Since K-Means++ is an unsupervised algorithm, internal clustering metrics were used instead of classification metrics. The following evaluation metrics were considered:

- **Silhouette Score** - measures separation between clusters (higher is better).
- **Davies–Bouldin Index (DBI)** – measures similarity between clusters (lower is better).
- **Calinski–Harabasz Score (CH Score)** - evaluates inter- and intra-cluster variance (higher is better).
- **Inertia (WCSS)** - measures cluster compactness (lower is better).

7.3.1 IoT Hardware Prototype Setup

To demonstrate the practical implementation of the proposed Infection Risk Detection System, an IoT hardware prototype was assembled using the ESP32 microcontroller, BME280, CCS811, and GP2Y1010AU0F dust sensor. The components were interconnected on a breadboard as shown in Figure 7.1. This prototype was used to collect real-time environmental data and transmit it to the ThingSpeak cloud platform for analysis.

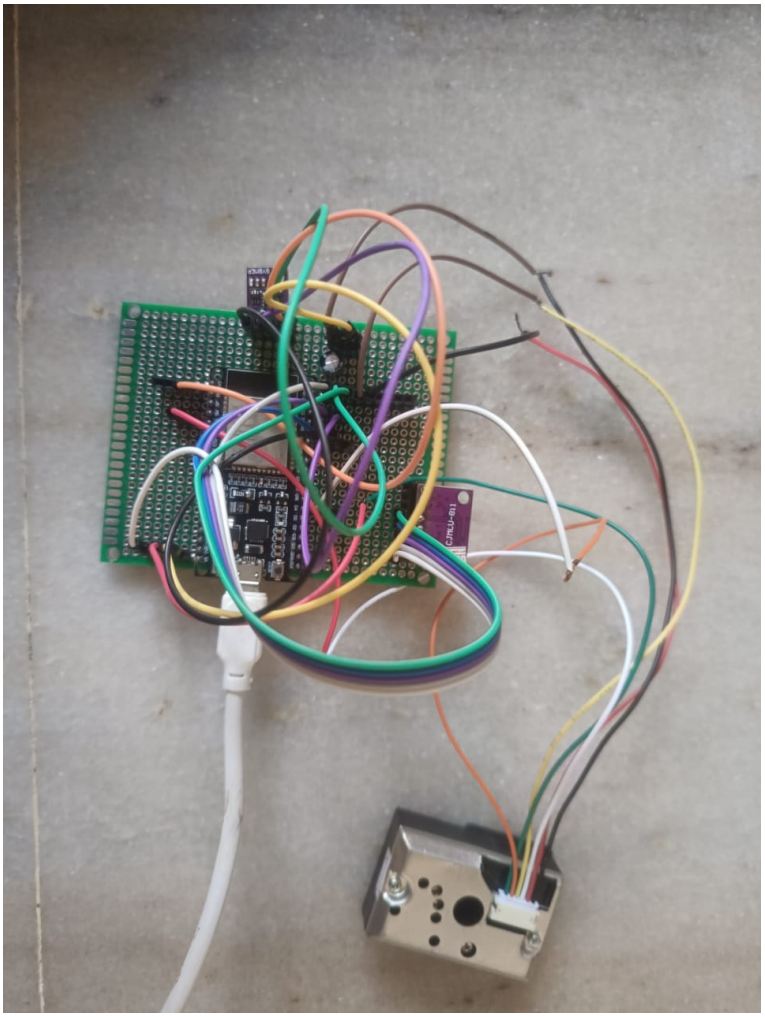


Figure 7.1: Prototype model of the device.

7.4 Infection Risk Prediction and Alert System

The selected PCA 2D K-Means++ model was deployed in the ThingSpeak cloud for real-time prediction. The ESP32 streams live sensor readings to the cloud, where the ML model classifies the environment into Low, Medium, or High risk. When the model predicts Medium or High risk, ThingSpeak triggers an automated telegram alert to health-care staff for timely intervention.

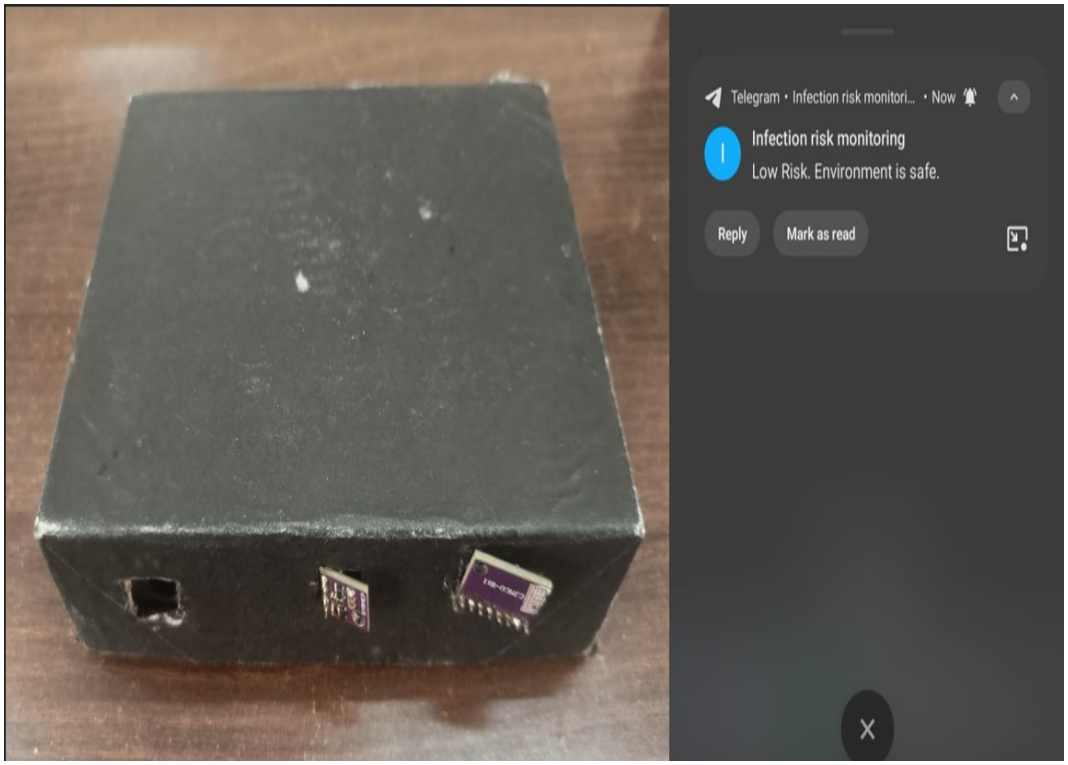


Figure 7.2: Real-time infection risk prediction alert.

This system ensures 24/7 automated monitoring of hospital environments, significantly reducing manual effort and enhancing infection prevention measures.Figure 7.2 shows the output i.e,device with alert notification.

Chapter 8

CONCLUSION AND FUTURE WORK

The proposed Infection Risk Detection System provides an effective solution for monitoring hospital indoor environments and identifying conditions that may lead to increased infection transmission. By integrating IoT-based sensing with machine learning, the system continuously observes critical environmental parameters such as temperature, humidity, air quality, and dust concentration. These parameters form the basis for predicting infection risk levels in real time.

The project demonstrates the successful implementation of an ESP32-based sensing system that collects environmental data from BME280, CCS811, and GP2Y1010AU0F sensors and uploads it to the ThingSpeak cloud platform. The K-Means++ clustering algorithm was employed to classify infection risk into Low, Medium, and High categories. Based on this classification, automated alerts are sent to hospital staff through telegram notifications, enabling timely preventive action such as improving ventilation, sanitization, or patient isolation.

The system achieved reliable performance, with K-Means++ producing high accuracy and stability compared to standard clustering techniques. This demonstrates the feasibility of using lightweight, unsupervised ML models in real-time healthcare monitoring applications. The continuous data transmission and cloud integration ensure that environmental conditions are monitored 24/7 without manual intervention, making the system suitable for critical hospital environments.

Although the system performs effectively in its current form, there are several promising directions for future work. One potential enhancement includes developing a dedicated mobile application for instant alerts, risk visualization, and remote monitoring by healthcare personnel. Future implementations may also incorporate additional sensors

such as CO_2-specific detectors or microbial air samplers to further improve risk prediction accuracy. Integrating supervised learning models could offer more precise and explainable results when larger labeled datasets become available.

The system could also be expanded to support multiple hospital rooms, enabling centralized monitoring of entire wards. Edge computing techniques may be integrated to enable offline prediction directly on the ESP32, reducing dependency on cloud platforms. Furthermore, enhancing the dashboard with predictive trends, anomaly detection, and maintenance alerts would significantly improve decision-making for hospital administration.

Overall, the proposed Infection Risk Detection System demonstrates a low-cost, scalable, and efficient approach to maintaining safer indoor environments in healthcare settings. With further improvements, it has the potential to evolve into a comprehensive infection prevention and environmental monitoring platform suitable for large-scale deployment.

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Appendix A

REPORT PLAGIARISM

Appendix B

PHOTO WITH CLIENT



Figure B.1: Photo with client

Contact Information

Name: Dr. B. SARITHA

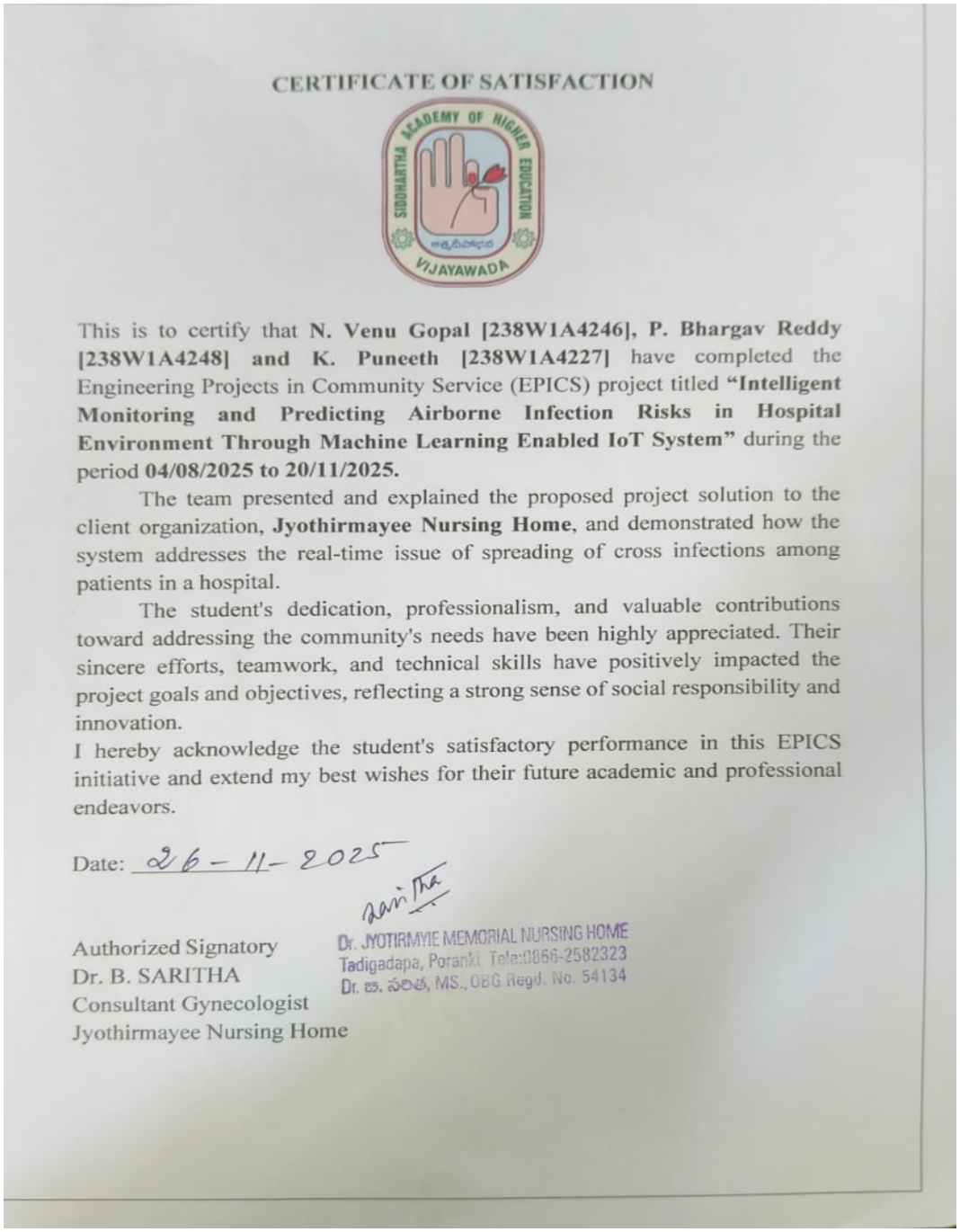
Profession: Doctor

Location: Poranki

Contact no. : +91 866-2582323

Appendix C

CLIENT SATISFACTION REPORT



CERTIFICATE OF SATISFACTION

This is to certify that **N. Venu Gopal [238W1A4246], P. Bhargav Reddy [238W1A4248] and K. Puneeth [238W1A4227]** have completed the Engineering Projects in Community Service (EPICS) project titled **"Intelligent Monitoring and Predicting Airborne Infection Risks in Hospital Environment Through Machine Learning Enabled IoT System"** during the period **04/08/2025 to 20/11/2025.**

The team presented and explained the proposed project solution to the client organization, **Jyothirmayee Nursing Home**, and demonstrated how the system addresses the real-time issue of spreading of cross infections among patients in a hospital.

The student's dedication, professionalism, and valuable contributions toward addressing the community's needs have been highly appreciated. Their sincere efforts, teamwork, and technical skills have positively impacted the project goals and objectives, reflecting a strong sense of social responsibility and innovation.

I hereby acknowledge the student's satisfactory performance in this EPICS initiative and extend my best wishes for their future academic and professional endeavors.

Date: 26-11-2025

Authorized Signatory
Dr. B. SARITHA
Consultant Gynecologist
Jyothirmayee Nursing Home

Dr. JYOTIRMYIE MEMORIAL NURSING HOME
Tadigadapa, Poranki Tele:0866-2582323
Dr. బ. సరిత, MS., OBG Regd. No. 54134

Figure C.1: Client satisfaction report